

L^AT_EX Exam Extras — Part B

The Discovery Catalogue
Features You May Not Know Exist

Read this one — do not reproduce it

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How to use this file. Do not try to reproduce it. Read it once now, and skim it again the night before. Its purpose is to remove the “I did not know that was a thing” problem: when a question paper shows an unfamiliar element, you need only recognise *which* tool makes it — the syntax is then one lookup away.

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1 Features you may not know exist

The catalogue is grouped by the kind of problem each tool solves. Every entry gives the package name, what it produces, and the smallest snippet that shows its syntax.

None of the packages catalogued here are loaded by this file, so the snippets are shown as code rather than rendered. That is deliberate: it keeps the file fast and lets it compile on a minimal installation.

1.1 Mathematics beyond the basics

Physics and vector calculus notation

The physics package turns painful constructions into one-word commands.

```
\usepackage{physics}
\dv{f}{x}           % df/dx as a proper derivative
\pdv[2]{f}{x}       % second partial derivative
\bra{\psi} \ket{\phi} \braket{\psi}{\phi} % Dirac notation
\abs{x} \norm{v}     % auto-sized bars
\grad \div \curl \laplacian
```

Auto-sized custom delimiters

mathtools lets you define a delimiter pair once and then get any size on demand — the professional replacement for typing `\left/\right` everywhere.

```
\usepackage{mathtools}
\DeclarePairedDelimiter{\abs}{\lvert}{\rvert}
\DeclarePairedDelimiter{\norm}{\lVert}{\rVert}
% usage: \abs{x} \abs*{\frac{a}{b}} \abs[\big]{y}
```

Also from mathtools: `\shortintertext{}` for a line of prose between aligned equations, `\mathclap` to stop wide subscripts spreading a formula, and `\coloneqq` for the definition symbol $\coloneqq$.

Boxed and highlighted equations

empheq emphasises a whole multi-line block, not just one line.

```
\usepackage{empheq}
\begin{empheq}[box=\fbox]{align}
  a \&= b + c \&\& d \&= e - f
\end{empheq}
\begin{empheq}[left=\empheqlbrace]{align} ... \end{empheq} % big brace
```

Chemistry

```
\usepackage[version=4]{mhchem}
\ce{H2O} \ce{SO4^2-} \ce{CO2 + C -> 2CO}
\ce{^{227}_{90}Th+}
\usepackage{chemfig} % structural diagrams
\chemfig{H-C(-[2]H)(-[6]H)-H}
```

1.2 Tables beyond the basics

Tables that break across pages

An ordinary `tabular` inside a `table` float cannot split. When a paper shows a table running over a page boundary with repeated headers, it is `longtable`.

```
\usepackage{longtable}
\begin{longtable}{|l|c|}
\caption{A long table.}\label{tab:long}\\
\hline Header & Value \\ \hline
\endfirsthead % header on the first page
\hline Header & Value \\ \hline
\endhead % header on every later page
\hline \multicolumn{2}{r}{\textit{continued}} \\
\endfoot
\hline
\endlastfoot
Row & 1 \\
% ... many rows ...
\end{longtable}
```

Note it is *not* placed inside a `table` environment — it provides its own caption mechanism.

Decimal alignment in a column

Numbers that line up on the decimal point look far more professional, and `siunitx` does it with the `S` column type.

```
\usepackage{siunitx}
\begin{tabular}{l S[table-format=3.2]}
\toprule
Sample & {\Mass (\si{\gram})} \\ \midrule
A & 1.20 \\
B & 143.75 \\
\bottomrule
\end{tabular}
% note: header text in an S column must be wrapped in braces
```

The older equivalent is `dcolumn`'s `\newcolumnntype{d}{D{.}{.}{2}}`.

Custom column types

Define once, reuse everywhere — especially a centred paragraph column, which plain $\text{\LaTeX}$ lacks.

```
\usepackage{array}
\newcolumnntype{C}[1]{>\centering\arraybackslash}p{#1}}
\newcolumnntype{R}[1]{>\raggedleft\arraybackslash}p{#1}}
\newcolumnntype{L}{>\bfseries}l      % a bold left column
% usage: \begin{tabular}{|L|C{3cm}|R{2cm}|}
```

Rescuing an oversized table

```
\resizebox{\textwidth}{!}{\begin{tabular}...\end{tabular}} % graphicx
\usepackage{adjustbox}
\begin{adjustbox}{max width=\textwidth} ... \end{adjustbox}
\small or \setlength{\tabcolsep}{3pt} % gentler alternatives
```

Modern all-in-one tables

`tabularray` replaces `tabular` + `multirow` + `colortbl` + `makecell` with a single key-value interface. Newer, so only use it if you know the exam machine has it.

```
\usepackage{tabularray}
\begin{tblr}{colspec={|l|c|c|}, row{1}={bg=blue!20, font=\bfseries}}
\hline Name & A & B \\ \hline
Row & 1 & 2 \\ \hline
\end{tblr}
```

1.3 Drawing things directly in $\text{\LaTeX}$

TikZ — the universal drawing package

If a paper shows a diagram that is clearly not a photograph — a flowchart, a block diagram, a state machine, a geometric construction — it was probably drawn with TikZ.

```
\usepackage{tikz}
\usetikzlibrary{shapes.geometric, arrows.meta, positioning}
\begin{tikzpicture}[node distance=2cm]
  \node (a) [rectangle, draw, rounded corners] {Start};
  \node (b) [ellipse, draw, below=of a] {Process};
  \draw [-Stealth] (a) -- (b) node[midway, right] {label};
\end{tikzpicture}
```

Circuit diagrams

```
\usepackage[siunitx]{circuitikz}
\begin{circuitikz}
  \draw (0,0) to[V, l=$v_{in}$] (0,3)
           to[R, l=$R$] (3,3)
           to[C, l=$C$] (3,0) -- (0,0);
\end{circuitikz}
```

Component keys include R, C, L, V, I, D (diode), `switch`, `short`; label with `l=`, current with `i=`, voltage with `v=`.

Plots from data or formulas

```
\usepackage{pgfplots}
\pgfplotsset{compat=1.18}
\begin{tikzpicture}
\begin{axis}[xlabel=$t$, ylabel=$v_C$, grid=major]
  \addplot[domain=0:5, samples=100, thick] {1 - exp(-x)};
  \addplot table[x=t, y=v] {data.dat}; % from a file
\end{axis}
\end{tikzpicture}
```

Trees, automata and specialised diagrams

```
\usepackage{forest} % syntax trees, decision trees
\begin{forest} [S [NP] [VP [V] [NP]]] \end{forest}

\usepackage{tikz}\usetikzlibrary{automata} % finite state machines
\usepackage{bytefield} % protocol/packet diagrams
\usepackage{qrcode} % \qrcode{https://example.com}
```

1.4 Algorithms and pseudocode

Two competing packages

```
% --- Option A: algorithm2e ---
\usepackage[ruled,vlined,linesnumbered]{algorithm2e}
\begin{algorithm}[H]
\KwIn{array $A$} \KwOut{maximum}
$m \gets A[1]$;
\For{$i \gets 2$ \KwTo $n$}{
  \If{$A[i] > m$}{
    $m \gets A[i]$;
  }
}
\Return $m$;
```

```

\caption{Find maximum}
\end{algorithm}

% --- Option B: algorithm + algpseudocode ---
\usepackage{algorithm}\usepackage{algpseudocode}
\begin{algorithm}
\caption{Find maximum}
\begin{algorithmic}[1]
\Procedure{Max}{$A, n$}
  \State $\mathit{m} \leftarrow A[1]$
  \For{$i \leftarrow 2$ \textbf{to} $n$}
    \If{$A[i] > m$} \State $\mathit{m} \leftarrow A[i]$ \EndIf
  \EndFor
  \State \Return $\mathit{m}$
\EndProcedure
\end{algorithmic}
\end{algorithm}

```

Tell them apart by looking at the output: `algorithm2e` uses `\;` line terminators and vertical block lines; `algpseudocode` uses explicit `\EndFor`/`\EndIf` keywords.

1.5 Page layout and furniture

Section heading appearance

```

\usepackage{titlesec}
\titleformat{\section}{\Large\bfseries\color{RoyalBlue}}{\thesection}{1em}{}
\titlespacing{\section}{0pt}{12pt}{6pt}
% also: \titleformat{\section}[block]{...}{...}{...}{...}[after-code]

```

Reach for this when headings in a target are coloured, centred, boxed, or have a rule under them.

Landscape pages and included PDFs

```

\usepackage{pdfscape} % rotates the page AND the PDF view
\begin{landscape} ... \end{landscape}

\usepackage{pdfpages} % paste in existing PDF pages
\includepdf[pages={1-3}]{appendix.pdf}
\includepdf[pages=-, nup=2x2]{slides.pdf}

```

Margin notes, epigraphs, watermarks

```

\usepackage{marginnote} \marginnote{A note in the margin.}
\usepackage{epigraph}   \epigraph{Quotation text.}{--- Author}
\usepackage{draftwatermark}
\SetWatermarkText{DRAFT}\SetWatermarkScale{4}
\usepackage{todonotes}   \todo{fix this later}

```

Changing geometry mid-document

```

\newgeometry{margin=0.5in} % new margins from here
... content ...
\restoregeometry           % back to the original

```

1.6 Text, typography and language

Quotation marks done properly

L^AT_EX quotes are asymmetric: ‘‘like this’’ gives “like this” (two backticks, two apostrophes). Straight double quotes produce wrong glyphs. The `csquotes` package automates it:

```

\usepackage{csquotes}
\enquote{quoted text} % correct marks for the document language
\blockquote{a longer passage}

```

Dashes: - hyphen, -- en-dash for ranges (pp. 10–20), --- em-dash for punctuation.

Underlining, striking and highlighting

```

\usepackage[normalem]{ulem}
\uline{underline} \uuline{double} \uwave{wavy} \sout{strike} \xout{crossed}
\usepackage{soul}
\hl{highlight} \ul{underline} \st{strike} \so{l e t t e r s p a c e d}
\sethlcolor{green} % change highlight colour

```

The `normalem` option is mandatory unless you want `\emph` silently converted to underlining throughout the document.

Micro-typography and spacing

```

\usepackage{microtype} % subtle margin kerning; just load it, no commands
\usepackage{setspace}  % \singlespacing \onehalfspacing \doublespacing
\usepackage{parskip}   % blank line between paragraphs, no indent
\raggedright           % turn off justification
\usepackage{xspace}    % for macros: \newcommand{\eg}{e.g. \@xspace}

```

Acronyms and glossaries

When a paper expands an abbreviation on first use and shortens it thereafter, that is automated:

```
\usepackage{acro}      % or glossaries / glossaries-extra
\DeclareAcronym{fft}{short=FFT, long=fast Fourier transform}
\ac{fft}                % first use: "fast Fourier transform (FFT)"; later: "FFT"
```

1.7 Structure, reuse and custom commands

Defining your own commands — the biggest time-saver

```
\newcommand{\R}{\mathbb{R}}                % no arguments
\newcommand{\dd}{\, , \mathrm{d}}
\newcommand{\pd}[2]{\frac{\partial #1}{\partial #2}} % two arguments
\newcommand{\vect}[1]{\bm{#1}}
\renewcommand{\arraystretch}{1.4}           % redefine an existing one
\newcommand{\greet}[2][Hello]{#1, #2!}     % #1 has default "Hello"

\newenvironment{note}                       % a custom environment
{\begin{quote}\itshape}\end{quote}}
```

If a symbol or construction appears more than three times in a paper, defining a macro for it in the preamble pays for itself immediately.

Splitting a document across files

```
\input{sections/intro}    % paste the file's contents here
\include{chapters/ch1}    % same, but forces a page break; chapters only
\includeonly{ch1,ch3}     % compile a subset while drafting
\usepackage{standalone}   % compile a figure alone or embedded
```

Conditionals and counters

```
\newcounter{mycount}\setcounter{mycount}{5}\stepcounter{mycount}
\themycount \arabic{mycount} \roman{mycount} \Alph{mycount}
\usepackage{ifthen}
\ifthenelse{\value{mycount}>3}{big}{small}
```

1.8 Referencing, deeper

Smarter references

```
\usepackage{cleveref}           % after hyperref
\cref{fig:a}                     % "fig. 1"
\Cref{fig:a}                     % "Fig. 1" (sentence start)
\cref{fig:a,fig:b,fig:c}         % "figs. 1, 2 and 3"
\crefname{figure}{Fig.}{Figs.}  % rename the type words
\crefrange{eq:1}{eq:5}          % "eqs. (1) to (5)"

\usepackage{varioref}
\vref{fig:a}                     % "figure 1 on the preceding page"

\nameref{sec:intro}              % the section's TITLE, not its number
```

PDF navigation

```
\usepackage{bookmark}           % better PDF bookmark control
\bookmarksetup{open=true, openlevel=2}
\hypersetup{bookmarksnumbered=true}
\phantomsection                 % fix a link target before \addcontentsline
```

`\phantomsection` before an unnumbered heading is the standard fix for a TOC link that jumps to the wrong place.

1.9 Bibliography, deeper

The three systems

```
% 1. Manual -- fastest in an exam, no extra passes
\begin{thebibliography}{9} \bibitem{k} ... \end{thebibliography}

% 2. BibTeX -- classic; compile pdflatex, bibtex, pdflatex, pdflatex
\bibliographystyle{ieeetr}      % plain, unsrt, abbrv, alpha, apalike, ieeetr
\bibliography{refs}

% 3. biblatex + biber -- modern, most flexible
\usepackage[backend=biber, style=numeric, sorting=none]{biblatex}
\addbibresource{refs.bib}
\printbibliography[title={References}]
```

Useful extras: `\nocite{*}` includes every entry in the `.bib` file whether cited or not; `natbib` adds author–year forms `\citet{}` (“Smith (2020) showed”) and `\citep{}` (“(Smith, 2020)”).

1.10 Small things that appear surprisingly often

Table 1: What to reach for when you recognise an element.

If the paper shows...	reach for
Coloured, titled panel of text	<code>tcolorbox</code>
Syntax-highlighted code	<code>listings</code> (or <code>minted</code>)
Pseudocode with keywords	<code>algorithm2e</code> / <code>algpseudocode</code>
Units set upright, aligned decimals	<code>siunitx</code>
A table crossing a page break	<code>longtable</code>
Notes under a table	<code>threeparttable</code>
Diagonally split header cell	<code>diagbox</code>
Line break inside a table cell	<code>makecell</code>
Clean rules, no vertical lines	<code>booktabs</code>
A flowchart or block diagram	<code>tikz</code>
A circuit schematic	<code>circuitikz</code>
A graph with axes and a grid	<code>pgfplots</code>
Chemical formulas	<code>mhchem</code>
Text wrapped around a figure	<code>wrapfig</code>
A rotated or landscape float	<code>rotating</code> / <code>pdflscape</code>
Panels labelled (a), (b), (c)	<code>subcaption</code>
An image forced exactly here	<code>float</code> and [H]
Highlighted or struck-through text	<code>soul</code> / <code>ulem</code>
Cancelled terms in a derivation	<code>cancel</code>
Bra-ket or derivative shorthand	<code>physics</code>
Custom bullet symbols, inline lists	<code>enumitem</code>
Running header with page number	<code>fancyhdr</code>
Coloured or restyled headings	<code>titlesec</code>
Two columns in part of a document	<code>multicol</code>
“Fig. 3” written automatically	<code>cleveref</code>
Acronyms expanded on first use	<code>acro</code> / <code>glossaries</code>
An index or glossary at the end	<code>makeidx</code> / <code>glossaries</code>
A QR code	<code>qrcode</code>
Existing PDF pages inserted	<code>pdfpages</code>
Dummy filler text	<code>lipsum</code> / <code>blindtext</code>

continued on next page

Table 1 continued

If the paper shows...	reach for
Placeholder images	mwe (example-image-a)

2 How to Read This Catalogue

1. **First pass — names only.** Read every box heading and the first sentence under it. Ignore the code. The goal is recognition, not recall.
2. **Second pass — the lookup table.** Study the table above. It is the fastest route from “the paper shows something odd” to “that is package X”.
3. **Third pass — your own shortlist.** Write down the six entries most likely to appear in your course. For an engineering paper that is usually `siunitx`, `listings`, `circuitikz`, `algorithm2e`, `booktabs` and `cleveref`.
4. **Night before.** Skim headings only. Five minutes.

Nothing here needs to be memorised as syntax. If you can name the tool, you can find the syntax; if you cannot name the tool, no amount of L^AT_EX skill will help you reproduce the element.

*You cannot reproduce a feature whose name you do not know.
Exercises and solutions are in `exam_extras_A.tex`.*